

AMAN KARN

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Dhulikhel, Kavre, Nepal

EDUCATION

- Kathmandu University, Nepal - Bachelor in Computer Engineering (5th Semester)
- +2 (Grade 12) - Completed Higher Secondary Education from St. Xavier's College, Maitighar with 3.92 GPA

TECHNICAL SKILLS

Python • JavaScript • TypeScript • C++ • React • Node.js • Express • Flask • Tailwind
CSS • MongoDB • MySQL • Firebase • Git • Postman • Vite • Machine Learning • NLP •
Data Analysis

PROJECTS

1. open IMIS Backend – NIYAM Module

- Developed a deterministic pre-submission validation module for the open IMIS backend using Python and Django.
- Integrated with the claim submission workflow to validate services, items, provider, policy, and attachment eligibility before claim processing.
- Designed a standalone validation engine producing ALLOW/WARN/BLOCK decisions with detailed reasoning and traceability.

2. Nepali Multimodal Fake News Dataset (NMFND)

- Built a toolkit to crawl Nepali news portals and synthesize fake news samples for multimodal AI research.
- Stored news articles, metadata, and images in SQLite using automated scraping and data pipelines.
- Created a structured multimodal dataset for fake news detection experiments.

3. Nepali Multimodal Fake News Detection

- Developed a multimodal fake news detection model combining MuRIL, CLIP, and MADF fusion.
- Implemented training, evaluation, and ablation pipelines using PyTorch and Hugging Face Transformers on Kaggle GPUs.

- Evaluated models using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

4. Nepali Multimodal Fake News Detection Web Application

- Built a full-stack deployment using React.js and FastAPI for real-time multimodal fake news inference.
- Integrated MuRIL, CLIP, and MADF models to classify Nepali news from text and images.
- Designed REST APIs and responsive frontend for research demonstration.

5. XiaomiHealthSync – Wearable Health Data Bridge (Project Proton)

- Built a React Native Android app that reads health metrics from Android Health Connect API (synced via Mi Fitness)
- Syncs heart rate, SpO2, steps, sleep, calories, and GPS location to Firebase Firestore every 15 minutes
- Runs as a lightweight background service on Android 14+ using Expo and Firebase SDK

6. Project Proton – Health-Aware Navigation System (Eco-thon 26)

- Built a health-aware navigation system combining smartwatch biometrics, ESP32 air quality sensors, and intelligent pathfinding
- Designed a risk score algorithm using heart rate, SpO2, PM2.5, and CO data to assign weighted map points
- Implemented Dijkstra/A* pathfinding on a weighted graph to route users through lowest-risk zones

7. ESP32 Air Quality Monitor – Firebase Bridge (Project Proton)

- Developed a Node.js bridge reading real-time air quality data from an ESP32 sensor over USB serial
- Captured dust density (GP2Y1010), CO levels (MQ-7), and air quality labels; uploaded to Firebase Firestore
- Maintained both latest-reading and historical Firestore collections for real-time and trend analysis

8. LaganiLens – NEPSE Trend Prediction System

- Built an end-to-end AI/ML pipeline for Nepal Stock Exchange analysis
- Scraped stock data from ShareSansar and stored in MongoDB
- Implemented ARIMA model for 30-day stock price forecasting
- Developed React-based frontend with portfolio management features

9. Nagarik Sewa – Digital Government Service Platform

- Developed a web platform simplifying access to government services
- Implemented eligibility checking, document tracking, and office mapping

- Provided step-by-step process guidance, cost, and time estimation
- Built using React, TypeScript, TailwindCSS

10. PeerMeet – Student Networking Platform

- Created a peer-to-peer networking system based on shared interests
- Used NLP embeddings for similarity-based recommendations
- Built backend with Node.js & Flask, frontend with React
- Integrated Firebase for authentication and database

ACHIEVEMENTS

- **Winner, HackforUHC 2026** – Developed the winning NIYAM backend validation module for openIMIS, improving deterministic pre-submission claim validation.
- **Speaker, openIMIS Community Meeting 2026** – Presented the NIYAM architecture, implementation, and validation workflow to the international openIMIS community.
- **Winner, SET Exhibition 2023, Technology category** – Designed a Bluetooth-enabled smart blind stick integrated with a mobile application, featuring voice-guided obstacle detection, emergency SOS calling, location sharing, and joystick-controlled accessibility functions for visually impaired users.
- **Top 100, National Maths Olympiad 2023**

LEADERSHIP & EXPERIENCE

- Research trainee at Health Informatics Lab, Kathmandu University
- Team Leader – NIYAM Project, HackforUHC 2026
- Founder – STEM Nepal
- Secretary – SXC Computer Club
- Online Presence Manager – The Crits
- Team Leader – Vision Project, SET Exhibition 2023
- Member – KU Computer Club